

Magnitude Without Symmetry

On the Misattribution of Cartwright-Type Bounds, the Closure Symmetry That Magnitude Presupposes, and a Symmetry-Indexed Generalization

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Status: Consolidated working note (v2.0), merging the prior critique (*A Critique of the False Application and Misattribution of Cartwright's Theorem*) and the generalization draft (*A Symmetry-Indexed Generalization of Cartwright's Theorem*) into a single document. Corrections from the prior versions are recorded explicitly in Section 0 and flagged inline where they occur.

0. Note on Corrections from the Prior Versions

This consolidation incorporates a verification that sharpens — and does not weaken — the central thesis. Three corrections are recorded here so the reader can see exactly what changed and why.

Correction 1 — The originating statement is itself a conflation, and this is now stated explicitly. The prior documents stated a bound of the form

$$|f(z)| \leq \frac{\max_{0 \leq i \leq p} |a_i|}{(1-r)^{2p}}, C_p$$

and referred to it, without qualification, as “Cartwright’s Theorem.” Verification against the literature shows that Mary Cartwright’s actual theorem is a result about **p-valent functions**: for every integer p there is a constant C_p such that every holomorphic function on the unit disk that attains no value more than p times satisfies a maximum-modulus bound with degradation $(1-r)^{-2p}$. The load-bearing hypothesis in Cartwright’s genuine theorem is **valence** — how many times a value is attained — *not* a constraint on the magnitude of Taylor coefficients. The expression stated above grafts a Cauchy-type coefficient bound (the max-coefficient numerator, the Taylor-coefficient control framing) onto Cartwright’s valence-derived $(1-r)^{-2p}$ degradation and constant C_p . **It is a fusion of two distinct results.**

This is not an error to be quietly removed. It is the *first rung of the very conflation this document diagnoses*, and it is now named as such. The misattribution begins at the level of the

theorem statement — before the symmetry conflation that the rest of the document analyzes even enters. We therefore distinguish throughout between (a) **Cartwright’s actual theorem** (valence-based), (b) the **Cauchy-type coefficient bound** (the genuine source of the max-coefficient numerator), and (c) the “**misuse form**” — the fused statement above, which we reproduce deliberately because it is the object whose misapplication we are studying. Where we write “Cartwright-type bound” without qualification, we mean the misuse form (c), and we mark it as such.

Correction 2 — The Lindemann, Euler, and Cartwright references are about misinterpretation by users, not about the mathematicians or their mathematics. This was the intent of the prior versions and is made unambiguous here: nothing in this document critiques Cartwright’s theorem, Lindemann’s proof, Euler’s identity, or their correct technical interpretations, all of which are sound and are stated carefully precisely so that the point about *conflation in misuse* is visible. The target is the inflation of frame-relative results into frame-independent ones by those who apply them — the act of conflation, named alongside the ways the results are *not* equivalent, so that the distinction carries meaning.

Correction 3 — The generalization is a framework, not a proved theorem, and is titled accordingly. The schematic “Symmetry-Indexed Cartwright-Type Bound” in Part B is a well-posed template with the classical case verified as a specialization; it is not a proved generalization theorem, because no nontrivial higher-symmetry instance is yet instantiated with a derived constant. It is presented as a framework and a research direction, with the conjectural rungs marked.

PART A — The Critique: Magnitude Presupposes Closure Symmetry

A.1. The Core Charge

The misuse form of the Cartwright-type bound is routinely invoked as a result *about magnitude* — bounding how large a function can become near a boundary given control over coefficient data. The charge is not against the mathematics, which (in each of its genuine components) is correct. The charge is against what the misuse treats the mathematics as being *about*, and what it silently presupposes in order to be about anything at all.

Every magnitude claim the misuse form produces is made possible by a prior **symmetry of closure** that it never names: the rotational symmetry of the unit disk's S^1 boundary. When the form is exported to settings where that closure symmetry is absent, weakened, or is itself the object of inquiry, the magnitude claims are not merely imprecise — they are *referentially ungrounded*. They compare without a stable term of comparison, measure without a fixed scale, and degrade without a well-defined direction of degradation.

The parallel to the common misreading of Lindemann is exact in structure. Lindemann proved that π is not the root of any nonzero polynomial with rational coefficients — a true and important result, correctly labeled “ π is transcendental” within that formal system. The *misinterpretation* layered on top — that π 's transcendence reflects an ontological gap between the circle and algebraic structure, that the circle is alien to algebra — inflates a formal-system-relative fact into an absolute claim. The circle is not irrational; it is the paradigm of closure symmetry. Its ratio to the linear diameter generates the artifact π , and that artifact is a property of the *comparison*, not of the circle. The misreading attributes to the circle a property that belongs to the act of measuring it with a linear instrument. The misuse of the Cartwright-type bound invites the structurally identical error: it attributes to functions in general a degradation law that actually belongs to a specific comparison performed within a specific symmetric domain.

A.2. What the Genuine Components Assume

State the components plainly before analyzing their misuse.

Cartwright's actual theorem controls maximum modulus via *valence*: a p -valent holomorphic function on $\mathcal{D}$ obeys a bound degrading as $(1-r)^{-2p}$. **The Cauchy-type coefficient estimate** controls a function's size via the modulus of its Taylor coefficients at the center. The **misuse form** fuses the coefficient numerator of the second onto the valence-derived degradation of the first. All three live on the same domain, and that domain is what matters here. It has four structural features:

1. **Rotational symmetry of the domain.** The boundary $\partial\mathcal{D} = S^1$ is homogeneous under $SO(2)$; every boundary point is equivalent to every other under rotation. No preferred direction, no distinguished point. This is the paradigm C-frame closed continuum.
2. **Radial distance as a scalar.** $r = |z|$ is a meaningful, direction-independent measure of proximity to the boundary *because* the boundary is equidistant from the center in every

direction. In an asymmetric domain, “distance from boundary” becomes direction-dependent and may fail to be a single scalar.

3. **Uniform (isotropic) degradation.** As $r \rightarrow 1$, $(1-r)^{2p}$ degrades at the same rate in every angular direction. This uniformity is a consequence of the boundary’s symmetry, not of any function f .
4. **A canonical center for expansion.** The expansion at 0 is privileged because the domain is symmetric about 0. Coefficients at an off-center point in a non-symmetric domain would not play the same structural role.

The conclusion: each genuine result, and a fortiori the misuse form, is *about* the symmetry of $\mathcal{D}$ — not merely *stated on* $\mathcal{D}$. The disk’s closure symmetry is the load-bearing structure that makes the terms (radial distance, uniform degradation, center-based coefficients) coherent. Remove the rotational symmetry and you do not get a weaker bound; you get a setting in which the bound’s terms lose their definiteness.

A.3. The False Application, in Three Types

Type 1 — Asymmetric degeneration boundaries. In configuration spaces with degeneration boundaries (points coinciding, collapsing, becoming singular), the “boundary” is typically a union of face strata of different codimensions and geometric characters, not equivalent under any group acting uniformly on all of them. Invoking a scalar $(1-r)^{2p}$ -type degradation presupposes a uniform rate the asymmetry cannot support; the true bound is direction-dependent in a way the misuse form conceals.

Type 2 — Magnitude of what, relative to what? The misuse form’s bound is meaningful on the disk because there is a fixed symmetric reference (the center, hence the coefficient maximum) and a fixed symmetric target ($\partial\mathcal{D}$). In an asymmetric setting, magnitude requires specifying: relative to which center? toward which boundary component? in which direction? Absent closure symmetry these have no canonical answers, so the magnitude claim floats free of any stable referent. The number refers to a measurement whose reference frame is dissolving.

Type 3 — Importing across encoding changes. The coefficients are data in a specific encoding (expansion at the center). Under re-coordinatization the coefficients transform, and $\max|a_i|$ in the new chart is not simply related to the old. A bound transported across charts without an explicit transport-operator correction carries the number but not the meaning. In

residue-groupoid terms this is a failure to account for the holonomy/loop-defect accumulated by cycling through encoding changes — exactly the quantity the misuse ignores.

A.4. The Lambda / NSAF / TUFT Diagnosis

Lambda. The degradation factor is the Lambda artifact: the irreducible residue of trying to control a C-frame holistic boundary object with L-frame sequential coefficient data. The misattribution reads this artifact as a property of the *function* rather than of the *comparison*. If it were a property of the function it would be portable; because it is a property of the comparison it is setting-specific, and exporting it to a different boundary geometry without re-deriving it is illegitimate.

NSAF. The unit-disk case sits at the simplest coupling: the boundary’s rotation action is free and uniform, so base data (coefficients) and fiber data (boundary phase) couple isotropically. Misapplying the form to a boundary whose strata carry different stabilizer structures is a **rung mismatch** — using a tool calibrated to the simplest symmetry level as though it were level-neutral. (*Caution, in the framework’s own anti-false-resonance spirit: the identification of the disk boundary S^1 with a Hopf fiber S^1 rests on the shared circle group and its rotation action, which is real, but the trivial fibration $S^1 \rightarrow S^1$ is degenerate; importing the full rung hierarchy from the NSAF shell ladder by name alone is not yet justified and should be proved or marked as analogy.*)

TUFT. The clean scalar $(1-r)^{2p}$ is a formal artifact of the boundary’s specific topological type (S^1 : compact, connected, homogeneous). A boundary of different type — stratified, cornered, non-compact, or of varying local topology — generates a different artifact that cannot be read off the formula. Treating all boundaries as S^1 -equivalent for boundary-approach control is precisely the collapse of formal boundary topology that TUFT is meant to flag, structurally identical to mistaking the boundary of the algebraic numbers for the boundary of mathematical reality in the Lindemann misreading.

A.5. Symmetry as the Precondition of Magnitude

The central philosophical charge, stated precisely: magnitude functions as a comparative term only given a fixed reference, a stable direction of comparison, and a closure that makes “near” and “far” mean the same thing to every observer. On the disk, all three are supplied by the S^1 symmetry — it closes the reference, stabilizes the direction, and universalizes the rate. Break

the symmetry and different directions of approach experience different degradation rates; the single scalar fractures into a direction-dependent function; and the magnitude has nowhere to stand. This is not a contingent fact about one theorem. It is the Lambda Principle at the level of measurement: magnitude is a C/L comparison artifact whose comparative stability depends on the closure symmetry of the object being measured. Symmetry is prior; magnitude is derivative.

PART B — A Symmetry-Indexed Generalization (Framework)

The critique implies a constructive program: parameterize the *validity* of a magnitude bound by symmetry class. The classical case is then the rung-1, S^1 -symmetric special case of a broader family. What follows is a framework with the classical case verified; the general theorem is schematic (see Correction 3).

B.1. Symmetry-Indexed Boundary Data

Replace the disk with a **boundary triple** $(X, \partial X, G)$: X a domain in $\mathbb{C}^n$ (or a complex manifold), ∂X its boundary, and G a group acting on ∂X by homeo-/diffeomorphisms. The **symmetry index** $\Sigma = [(\partial X, G)]$ is the equivalence class of the boundary-symmetry pair under conjugacy of actions (or coarse geometric equivalence). Σ encodes how homogeneous and how isotropic the boundary is, and different Σ support different kinds of magnitude statement. The classical case is recovered as $X = \mathcal{D}$, $\partial X = S^1$, $G = SO(2)$.

B.2. Encodings and the C/L Comparison

A C-frame object is f as a section over X with boundary behavior on ∂X ; an L-frame encoding $E: \mathcal{F}(X) \rightarrow \mathcal{L}$ maps it to coefficient data (Taylor at a point, Fourier along a boundary component, or eigenfunction expansion of a Laplacian adapted to $(X, \partial X, G)$). The generalized Cartwright-type question is: *given Σ and E , what constraints on $E(f)$ are forced by controlled boundary behavior of f near ∂X ?*

B.3. Rung and Residue Structure

At **rung 1** the coupling between encoding (base) and boundary behavior (fiber) is homogeneous; the residue groupoid $\mathcal{R}$ capturing how local boundary data glue is essentially trivial, and magnitude transport across the boundary is path-independent. At **higher rungs** $\mathcal{R}$ may carry nontrivial holonomy and magnitude comparison becomes path-dependent or stratified. A Cartwright-type bound is meaningful only when $\mathcal{R}$ is tame (rung 1, or rung-k with an explicit correction operator).

B.4. The Schematic Theorem (Template, Not Yet Proved)

Template (Symmetry-Indexed Cartwright-Type Bound). Let $(X, \partial X, G, \Sigma)$, $\mathcal{F}(X)$, E , and a symmetry-adapted boundary distance ρ satisfy: (1) symmetry-compatible approach regions $A_\epsilon = \{x : 0 < \rho(x) < \epsilon\}$ with no privileged direction within a G -orbit; (2) rung-1 residue, or rung-k with an explicit transport correction T_Σ ; (3) an encoding adapted to Σ with compatible norms. If f obeys a boundary growth condition $\sup_{A_\epsilon} \Psi_\Sigma(f) \leq M_\Sigma(\epsilon)$, then there is a degradation operator D_Σ on $\mathcal{L}$ and a constant K_Σ (depending only on Σ , the rung, and the encoding) with

$$\|D_\Sigma(E(f))\|_{\mathcal{L}} \leq K_\Sigma \cdot \limsup_{\epsilon \rightarrow 0^+} M_\Sigma(\epsilon).$$

Classical specialization (verified). $\Sigma = [(S^1, SO(2))]$, $\rho = 1 - |z|$, $\Psi_\Sigma = |f|$, $M_\Sigma(\epsilon) \sim \epsilon^{-2p}$, D_Σ acting as multiplication by $(1-r)^{2p}$; the misuse form is recovered as one specific choice of $(\Sigma, \rho, \Psi_\Sigma, D_\Sigma)$. This recovery is a correct consistency check; it does not by itself establish any nontrivial instance.

B.5. Symmetry Classes and Their Degradation Profiles

- **Homogeneous isotropic (Cartwright-like):** scalar profile $M_\Sigma(\epsilon) \sim \epsilon^{-\alpha_\Sigma}$, $D_\Sigma \sim \rho^{\alpha_\Sigma}$. Direct generalization of the disk.
- **Stratified:** $\partial X = \sqcup S_j$ with possibly different G_j ; profiles become piecewise and D_Σ block-structured. Magnitude bounds are stratified, not global — this is where naive application fails.

- **Anisotropic:** non-isotropic G-action gives direction-dependent ρ and a tensorial profile $M_\Sigma(\epsilon, \omega) \sim \epsilon^\alpha \{-\alpha(\omega)\}$. Here any scalar Cartwright-type bound is structurally invalid; only orbit-resolved bounds make sense.

B.6. Worked Direction: Pythagorean Interval Lattice (Conjectural Instance)

A concrete higher-structure target: take the 2–3–5 just-intonation lattice Λ as the L-frame, with a resonance sublattice Λ_{res} (octaves, fifths, just chords) and a resonance distance ρ_{Pyth} combining exponent displacement and log-frequency displacement, vanishing exactly on Λ_{res} . A degradation operator $(D_\Sigma A)(a,b,c) = \rho_{\text{Pyth}}(a,b,c)^\gamma \cdot A(a,b,c)$ then expresses a Cartwright-type constraint: large coefficients allowed only near resonance. Chords define sublattices with their own Σ_{chord} ; nested grids (2–3, then 2–3–5, then tempered, then extended primes) form an NSAF layer structure, with *transitional harmonics* the modes that remain near-resonant across projections between layers, i.e. where degradation operators approximately commute with the inter-layer projection.

Status of this instance. The objects $(\rho_{\text{Pyth}}, D_\Sigma \text{Pyth}, \text{the layer projections})$ are *defined*; the Cartwright-type bound $\|D_\Sigma A\| \leq K$ is *posited*, with K not yet derived from the lattice structure. This is the most developed candidate for a nontrivial instance, but until K is constructed it remains conjectural — a well-posed target, not a proved bound. It is included to show what instantiation would require, and to mark precisely the line where the framework currently stops.

C. Summary and Honest Status

The critique (Part A) is the load-bearing result and stands on its own: magnitude is a C/L comparison artifact whose stability is borrowed from a closure symmetry, and exporting a magnitude bound across a change of boundary symmetry — without an explicit transport correction — produces numbers without referential ground. This holds independently of any framework and is the part a working analyst can check directly. The Lindemann parallel is structural and exact: a frame-relative result inflated into a frame-independent one.

The generalization (Part B) is a well-posed framework that correctly recovers the classical case as its rung-1 specialization, correctly identifies stratified and anisotropic boundaries as the

regimes where scalar bounds fail, and supplies definitions for higher-rung degradation operators. It is not yet a proved generalization theorem; the higher-rung instances, including the Pythagorean-lattice candidate, are defined but not yet instantiated with a derived constant, and are marked conjectural.

The corrections in Section 0 strengthen rather than weaken the thesis: the discovery that the originating “Cartwright” statement is itself a fusion of Cartwright’s valence theorem with a Cauchy-type coefficient bound is not a flaw in the argument — it is the first and clearest instance of the conflation the document exists to diagnose, occurring at the level of the theorem’s name before any symmetry question is raised. Naming it there, plainly, alongside the ways the genuine results are *not* equivalent, is what gives the distinction its meaning.

This is a consolidated working note. The critique is offered as a sound structural argument; the generalization as a framework and research direction with its proved and conjectural parts explicitly separated. The mathematics of Cartwright, Lindemann, Euler, and the Cauchy estimate is not in dispute; only its conflation in misuse is.